Where the Predictability Is: Conditional Variance, Not Conditional Mean, Delivers Calibrated Daily Return Forecasts for Large-Cap Technology Stocks, 2016–2025

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Abstract

We ask where the predictability in daily large-capitalisation technology equity returns resides, using a panel of eight United States stocks over 2016–2025 and a held-out window spanning the 2022–2023 monetary tightening cycle. Six model classes—baselines, ridge, random forests, GARCH-family volatility models, four sequence architectures and a hierarchical Bayesian regression with a horseshoe prior—are estimated on one macro-financial information set and scored on point accuracy, probabilistic accuracy and interval calibration, with every comparison tested by Diebold–Mariano. We find no conditional-mean predictability: zero of 48 model–ticker tests survive multiplicity correction, and when the sequence models are permitted to select their own regularisation on held-out data all four discard the predictors and forecast a constant. The conditional variance behaves differently: all three GARCH specifications beat a rolling-variance benchmark at $p < 0.001$. Decomposing the gain in log predictive density over a null forecast, the conditional variance contributes 54% and the shape of the predictive distribution a further 45%, while the conditional mean contributes 0.7%—statistically detectable at $p = 0.029$ and economically negligible. Student-t innovations with a validation-calibrated scale reduce mean absolute coverage error to 0.022, and partial pooling yields usable intervals for an asset with no estimation history. We also document a silent failure mode: standardising non-stationary macroeconomic levels on a fixed origin sends held-out inputs to 8.7 training standard deviations and produces out-of-sample R^2 as low as $-10\,425\%$ with no error raised. All code, audits and pinned dependencies are released.

Keywords: return predictability; forecast calibration; GARCH; hierarchical Bayesian models; distribution shift

1 Introduction

Between 2016 and 2025, large-capitalisation United States technology equities traded through several distinct macro-financial regimes: an extended period of near-zero policy rates, the COVID-19 market dislocation of March 2020, the most rapid monetary tightening cycle since the early 1980s beginning in 2022, and the partial normalisation of rates that followed. These episodes produced pronounced volatility clustering, heavy-tailed daily returns, and repeated shifts in the joint distribution of returns and macroeconomic state variables—features long documented as stylised facts of asset returns [Cont, 2001]. Such conditions are hostile to forecasting models estimated under an assumption of covariance stationarity, and they reproduce at the level of a portfolio a persistent finding of the empirical finance literature: predictive relations that appear stable in sample frequently deteriorate out of sample [Welch and Goyal, 2008, Timmermann, 2008]. As West [2020] argues, structure uncertainty then becomes as consequential as parameter uncertainty, since the analyst must decide not only what the coefficients are but which model, if any, remains adequate as the regime changes.

There are sound theoretical grounds for conditioning return forecasts on macro-financial information. Multifactor asset pricing theory implies that innovations in systematic state variables are priced in equity returns [Chen et al., 1986]. Empirically, unanticipated changes in the federal funds rate induce large and statistically significant equity market responses [Bernanke and Kuttner, 2005]; option-implied volatility, summarised by the CBOE Volatility Index, is contemporaneously and negatively associated with returns [Whaley, 2000]; and comovement with the broad market index remains the dominant source of systematic variation in individual stock returns. Recent data-driven studies confirm that volatility and uncertainty measures rank among the most influential predictors of index returns [Latif et al., 2025], and that volatility transmission across markets is itself time varying [Agrawal et al., 2026]. The unresolved methodological question is therefore not whether such variables carry information, but how they should enter the forecasting model and how the resulting predictive uncertainty should be quantified.

This paper addresses that question by decomposing the problem rather than by running a horse race. A predictive distribution for a daily return has a location, a scale and a shape, and a forecaster can invest effort in any of the three. We ask which investment pays.

The design is a controlled comparison on a balanced panel of eight large-cap technology stocks over 2016–2025, with the effective federal funds rate, the ten-year Treasury yield, the VIX and a market proxy serving as macro-financial conditioning information. Six model classes are estimated on one identical feature set: naive baselines, pooled ridge, a pooled random forest, three GARCH-family volatility specifications, four sequence architectures, and a hierarchical Bayesian regression that treats the GARCH conditional standard deviation as a known observation scale and places a horseshoe prior on the mean coefficients. All are fitted on a chronological training sample and evaluated on a strictly held-out window spanning the tightening cycle and its aftermath, so that out-of-sample performance is measured across materially different volatility regimes. Point accuracy is assessed by root mean squared error and out-of-sample R^2 against a per-ticker mean benchmark; probabilistic adequacy by logarithmic score, continuous ranked probability score and the calibration of predictive intervals. Every comparison is tested by Diebold–Mariano rather than ranked.

Two features of the problem motivate this design. First, at the daily horizon the signal-to-noise ratio in equity returns is low, and differences in point accuracy across competing model classes are typically negligible; an evaluation confined to point forecasts therefore yields an incomplete and potentially misleading ordering. Second, the model classes that dominate the recent forecasting literature—tree ensembles and recurrent neural networks—produce predictions without an accompanying probability model, and so cannot state how much confidence a forecast warrants or how that confidence should contract when the market moves from a calm to a turbulent state [Chandra and He, 2021, Wong et al., 2024].

Our findings are largely negative on the first moment and positive on the second. No model class improves on a per-ticker historical average, and the result survives disaggregation to individual assets and correction for multiple testing; when the sequence models are permitted to select their own regularisation on held-out data, all four discard the predictors entirely and forecast a constant. The conditional variance behaves differently, with all three GARCH specifications beating a rolling-variance benchmark decisively. Decomposing the gain in log predictive density over a null forecast, the conditional variance contributes 54% and the shape of the predictive distribution a further 45%, while the conditional mean contributes 0.7%.

We also document a failure mode that is silent rather than merely severe. Standardising non-stationary macroeconomic levels on a fixed origin sends held-out inputs far outside the range seen in estimation, and every model with an unbounded output head extrapolates along them, producing out-of-sample R^2 as low as $-10\,425\%$ with no exception raised and no implausible training diagnostic. Because the defect is detectable only by auditing predictor distributions across splits and the scale of the predictions themselves, both audits are released with the code.

The remainder of the paper reviews the relevant literature, describes the data and methodology, reports the empirical results, and concludes with recommendations for practice.

2 Literature Review

2.1 Classical time-series models and the limits of return predictability

The autoregressive integrated moving average framework of Box and Jenkins [1970] remains the canonical linear model for the conditional mean of a univariate series and the natural benchmark against which richer specifications are judged. Its known limitation in financial applications lies in the second moment: daily returns exhibit conditional heteroskedasticity, which motivated the ARCH model of Engle [1982] and its GARCH generalisation [Bollerslev, 1986], and which implies that a correctly specified linear conditional-mean model may still misstate predictive uncertainty by a wide margin. A second limitation concerns the conditional mean itself. In a comprehensive out-of-sample audit, Welch and Goyal [2008] found that most proposed return predictors fail to improve upon the historical average, and Timmermann [2008] characterised return predictability as local and episodic—present in short-lived pockets rather than as a stable population property. Forecast combination offers partial insurance against this instability [Rapach et al., 2010], but the collective evidence establishes a demanding baseline: any gain claimed for a more elaborate model must survive genuine out-of-sample evaluation over heterogeneous regimes.

2.2 Machine learning approaches

Machine learning methods relax the linearity restriction at the cost of a larger effective parameter space. Random forests [Breiman, 2001] control overfitting through bootstrap aggregation of decorrelated trees and provide internal measures of predictor importance, properties that make them a common nonlinear benchmark in return forecasting. In the most systematic comparison to date, Gu et al. [2020] showed that regression trees and neural networks yield economically significant improvements in measuring equity risk premia, attributing the gains chiefly to nonlinear predictor interactions that linear methods miss. For sequence models, the LSTM architecture [Hochreiter and Schmidhuber, 1997] was shown by Fischer and Krauss [2018] to outperform random forests and logistic regression in classifying daily S&P 500 constituent returns, although the documented profitability of the strategy declined in later subperiods—an instability consistent with the pocketed predictability described above. Subsequent applied work has extended these architectures with macroeconomic inputs and hybrid convolutional–recurrent designs [Latif et al., 2025]. Two limitations recur throughout this literature: the fitted models are difficult to interpret, and their outputs are point predictions unaccompanied by a distributional statement.

A third limitation is less often discussed and is central to this paper. Because these models are typically reported at a single regularisation setting chosen by the author rather than selected on held-out data, it is difficult to distinguish a model that has found signal from one that has fitted noise. Section 4.2 shows that the distinction is decisive here.

2.3 Bayesian inference and uncertainty quantification

The Bayesian literature addresses the distributional limitation directly. West [2020] surveys multivariate Bayesian forecasting with emphasis on scalability and structure uncertainty, arguing that decision-relevant forecasts require the full predictive distribution rather than a summary of its centre. Qiu et al. [2018] develop the multivariate Bayesian structural time series model, in which local linear trends, seasonal components and spike-and-slab selection over a candidate pool of predictors jointly capture contemporaneous correlation across target series while guarding against overfitting. In the return-predictability setting specifically, Bayesian treatments of model uncertainty—averaging over predictor subsets rather than conditioning on a single specification—were shown by Avramov [2002] and Cremers [2002] to improve out-of-sample performance precisely because no individual model is reliably correct. Closer to the present study, Chandra and He [2021] apply Bayesian neural networks to multi-step stock price forecasting and report usable predictive intervals even through the COVID-19 volatility spike, while Wong et al. [2024] show that standard uncertainty-quantification schemes for neural networks miscalibrate under volatility clustering, underscoring that calibration must be verified rather than assumed. Proper scoring rules [Gneiting and Raftery, 2007] and tests of equal predictive accuracy [Diebold and Mariano, 1995] supply the formal apparatus for such verification, and this paper applies both throughout.

For coefficient shrinkage we adopt the horseshoe prior of Carvalho et al. [2010], the continuous analogue of the spike-and-slab selection used by Qiu et al. [2018], sampled by the auxiliary-variable scheme of Makalic and Schmidt [2016].

2.4 Distribution shift and the fragility of fixed-origin evaluation

A strand of the machine learning literature that has had limited uptake in empirical finance concerns dataset shift: the case in which the joint distribution of predictors and target differs between estimation and deployment [Quiñonero-Candela et al., 2009]. The concern is directly relevant here, because the macro-financial variables most commonly used to condition equity forecasts—an index level, a policy rate, a yield—are precisely the series whose means move permanently across a monetary regime change. Standardising such a series on a fixed estimation window and applying the transformation to a later window produces inputs far outside the range the estimator has seen, and any model with an unbounded output head extrapolates along them. Section 4.8 documents the consequences and the repair.

2.5 Hybrid architectures and the remaining gap

A growing strand seeks to combine the representational capacity of recurrent networks with the distributional discipline of econometric volatility models. Kim and Won [2018] integrate LSTM networks with multiple GARCH-type specifications and report improved index volatility forecasts relative to either component alone, and Wang et al. [2022] document analogous gains for a combined LSTM–GARCH-family model in a highly volatile price series. These hybrids, however, typically attach a volatility filter to a point-forecasting network rather than deliver a coherent posterior predictive distribution, and published comparisons rarely hold the information set fixed across model families.

The present study is positioned in that gap. It evaluates classical, machine learning and Bayesian models on a single panel with an identical macro-financial predictor set and an identical regime-spanning out-of-sample window, and it scores them on both point accuracy and the calibration of their stated uncertainty. To the extent that the literature reviewed above establishes weak and unstable conditional-mean predictability alongside strong and persistent structure in conditional variances, the comparison speaks to a practical question left open by prior work: whether the principal contribution of macroeconomic drivers and Bayesian inference at the daily horizon lies in sharper point forecasts or in more honest predictive distributions. The decomposition in Section 4.4 answers it quantitatively.

3 Data and Methodology

3.1 Sample and splits

The sample is a balanced daily panel of eight large-capitalisation United States technology equities—AAPL, MSFT, AMZN, GOOGL, META, NVDA, TSLA and ORCL—from 2 March 2016 to 29 December 2025. Adjusted closing prices are obtained from Yahoo Finance and macro-financial series from the Federal Reserve Economic Data service: the effective federal funds rate (DFF), the ten-year Treasury constant maturity yield (DGS10), the S&P 500 index (SP500) and the CBOE volatility index (VIX). The market proxy is the SPY exchange-traded fund.

The forecast target is the one-day-ahead log return. For ticker i, a panel row is keyed by date t and its target is

$$y_{i,t} = \log P_{i,t+1} - \log P_{i,t}, \tag{1}$$

realised on $t+1$. Every predictor is observable at the close of t, so no input uses information from the period being forecast.

The panel is divided chronologically into three blocks, with no shuffling at any stage:

Split	Dates	Rows
Training	2016-03-02 to 2019-12-30	7 720
Validation	2019-12-31 to 2022-12-29	6 048
Test	2022-12-30 to 2025-12-29	6 008

The split is deliberately demanding. The training block spans a low-rate, low-volatility regime; the validation block contains the COVID-19 dislocation; and the test block spans the 2022–2023 tightening cycle and its aftermath. Any relation that survives is required to survive a change in monetary regime.

3.2 Feature dictionary and stationarity

Model inputs are declared explicitly rather than inferred from column type. This is a substantive design choice, not bookkeeping. Selecting every numeric column admits level series—an index level, a policy rate—whose mean moves permanently between the training and test blocks. Standardised on training moments and applied to the test block, such a series arrives at a magnitude the estimator has never seen, and any model with an unbounded linear output extrapolates along it. Section 4.8 reports the consequences directly.

Membership is therefore fixed in advance, and each candidate is classified by an augmented Dickey–Fuller test. Three series are non-stationary over the sample: the S&P 500 level ($p = 0.99$), the effective federal funds rate ($p = 0.83$) and the ten-year yield ($p = 0.80$). All three are excluded and replaced by their first differences or by a stationary counterpart. The retained set comprises fourteen predictors: six own-return lags, three trailing realised volatilities, the market return, the log VIX and its change, and the daily changes in the federal funds rate and the ten-year yield. Hybrid specifications add one further input, the GARCH conditional standard deviation of Section 3.4, so base and hybrid models differ by exactly one column. Table 1 lists every candidate, retained or excluded, with the reason for exclusion.

Two redundancies are also removed. One column duplicated another exactly, and the S&P 500 return correlated 0.9985 with the market proxy return; retaining both would inflate collinearity and render the Bayesian coefficient posteriors uninterpretable.

3.3 Evaluation protocols

Two protocols are used throughout, and the difference between them is itself a result.

Under the *fixed-origin* protocol a model is estimated once on the training block, and the fitted model together with its feature scaler is frozen and applied to the whole test block. This is the conventional design and is appropriate when predictors are stationary.

Under the *expanding-window* protocol the model walks forward month by month. At each fold it is refitted on every observation whose target date falls strictly before the fold start, and it forecasts that month only; the scaler is refitted inside the fold along with the model. This is what a forecaster could have executed in real time.

Both protocols score the identical 6 008 test observations, so their metrics are directly comparable, and across all 36 folds no training observation reaches into the month being scored.

3.4 Conditional volatility

Three conditional volatility specifications are estimated per ticker on returns in percent: GARCH(1,1), GJR-GARCH(1,1) and EGARCH(1,1), each with Student-t innovations. Parameters are estimated on the training block alone. They are then held fixed and the recursion is filtered forward through the validation and test blocks, so the stored quantity is a genuine one-step-ahead forecast $\sigma_{i,t+1|t}$ that uses only information through t. The value recorded at date t therefore forecasts the same period as the target in (1).

Every fit is checked for optimiser convergence, finite parameters and a conditional standard deviation within a plausible range before it is written; the pipeline aborts otherwise. All twenty-four fits (eight tickers, three specifications) satisfy these conditions, with conditional standard deviations between 0.6% and 13.8% per day against a realised return standard deviation of 2.5%. Per-fit parameters, log-likelihoods and AIC values appear in Appendix Table **??**.

3.5 Hierarchical Bayesian model

The central specification pools information across tickers while treating the GARCH forecast as a known observation-level scale. For ticker i on day t, with $\mathbf{x}_{i,t}$ the standardised predictor vector,

$$y_{i,t} \mid \alpha_i, \boldsymbol{\beta}, s^2 \sim t_\nu\Big(\alpha_i + \mathbf{x}_{i,t}^\top \boldsymbol{\beta},\ s^2 \sigma^2_{i,t+1|t}\Big), \tag{2}$$

$$\alpha_i \mid \mu_\alpha, \tau_\alpha^2 \sim \mathcal{N}(\mu_\alpha, \tau_\alpha^2), \tag{3}$$

$$\beta_j \mid \lambda_j, \tau_\beta \sim \mathcal{N}(0, \lambda_j^2 \tau_\beta^2), \qquad \lambda_j \sim \mathcal{C}^+(0,1), \quad \tau_\beta \sim \mathcal{C}^+(0,1). \tag{4}$$

Three features of this specification carry the analysis.

The GARCH forecast enters (2) as a known scale, so the model does not relearn volatility clustering and s^2 is a single scalar measuring whether that scale is correctly sized overall. A value near one means the volatility model is already calibrated.

Equation (3) is the partial-pooling layer. Because α_i is drawn from a population distribution, a ticker absent from the training sample can still be forecast by integrating α over that distribution, which adds τ_α^2 to the predictive variance. Section 4.7 exploits this. Following Gelman [2006], τ_α receives a half-Cauchy prior rather than a conventional inverse-gamma: with only eight groups the between-ticker standard deviation is weakly identified, and a diffuse inverse-gamma prior largely reproduces its own mode.

Equation (4) is the horseshoe prior of Carvalho et al. [2010]. It replaces the more common practice of pre-screening predictors by their correlation with the target. That practice is unsafe in this setting for a specific reason: a trending series correlates with almost any target over a multi-year window, so a correlation screen preferentially selects exactly the non-stationary levels that Section 3.2 excludes. The horseshoe retains every predictor and shrinks weak coefficients continuously toward zero while leaving strong ones nearly unshrunk, so selection occurs inside the posterior and carries uncertainty. It is the continuous analogue of the spike-and-slab selection of Qiu et al. [2018].

Student-t innovations are represented as a scale mixture of normals, $y \sim \mathcal{N}(\mu, s^2\sigma^2/\omega)$ with $\omega \sim \text{Gamma}(\nu/2, \nu/2)$, which recovers (2) marginally and keeps every conditional in closed form.

Inference is by blocked Gibbs sampling. All conditionals are conjugate, including the horseshoe scales, via the auxiliary-variable representation of Makalic and Schmidt [2016]; the degrees of freedom ν are drawn by griddy Gibbs. Four chains are run from dispersed starting points for 40 000 iterations, discarding the first 20 000 and thinning by four, giving 20 000 retained draws. Convergence is assessed by rank-normalised $\widehat{R}$ and effective sample size [Vehtari et al., 2021]: across 27 monitored quantities the largest $\widehat{R}$ is 1.0033 and the smallest bulk effective sample size is 1 703.

The sampler is validated on synthetic data in which three of twelve coefficients are non-zero. It recovers all three signal coefficients with credible intervals excluding zero, produces no false positives among the nine null coefficients, and recovers the observation scale and the between-group standard deviation.

3.6 Predictive distributions and scale calibration

Predictive means and scales use the full posterior rather than plug-in point estimates, so parameter uncertainty enters the interval through the law of total variance.

One calibration step is applied. GARCH parameters are frozen at the end of the training block, and the training block is calmer than what follows: measured as the standard deviation of y/σ, the scale the data require is 1.002 on the training block but 1.118 on validation and 1.117 on test. A scale fitted to training data alone therefore understates later volatility by roughly twelve percent. A single scalar is estimated on the *validation* block and applied to the test block; no test observation informs it.

The matching statistic matters. Because the validation block contains the COVID-19 dislocation, a variance match is dominated by a few extreme days, which is behaviour the Student-t tails are meant to absorb rather than the scale. Matching the interquartile range targets the body of the distribution, which is what the scale parameter governs, and yields a factor of 1.195 against 1.220 for the variance match.

3.7 Scoring rules and inference

Point accuracy is reported as root mean squared error, mean absolute error, directional accuracy and out-of-sample R^2 against a stated reference forecast. The reference is the per-ticker mean computed under the same protocol as the model being scored, so it never enjoys an information advantage.

Probabilistic accuracy is reported as the mean logarithmic score, the continuous ranked probability score and empirical coverage at the 50%, 80%, 90% and 95% levels [Gneiting and Raftery, 2007]. All three are reported because they can disagree: a model may be better calibrated while being less sharp. Calibration is assessed additionally by the probability integral transform, whose values are uniform under a correctly specified predictive distribution, tested formally by a Kolmogorov–Smirnov statistic.

Volatility forecasts are compared under QLIKE against the squared return as the realised-variance proxy. QLIKE is chosen because it preserves the ranking a perfect proxy would give even when the available proxy is noisy [Patton, 2011].

All comparisons are tested rather than asserted. Equal predictive accuracy is assessed by the Diebold and Mariano [1995] test on per-observation loss differentials, using a Newey–West heteroskedasticity- and autocorrelation-consistent variance because daily loss differentials are serially correlated, together with the small-sample correction of Harvey et al. [1997]. Where a family of tests is reported—one per model–ticker pair—significance is additionally assessed after Benjamini–Hochberg and Bonferroni correction, since at conventional levels a handful of rejections among dozens of tests is what chance alone produces.

4 Results and Discussion

All results are computed on the same held-out block of 6 008 observations covering 2023–2025, against the same per-ticker mean benchmark, and every comparison is tested rather than asserted.

4.1 The conditional mean

No model class improves on the per-ticker mean benchmark. Under the fixed-origin protocol, pooled ridge attains an out-of-sample R^2 of -0.11% and the pooled random forest -0.34%; under the expanding-window protocol the corresponding figures are -0.28% and -0.31%. Diebold–Mariano tests against the benchmark return p-values of 0.36, 0.18, 0.17 and 0.25 respectively. None of the four is distinguishable from a forecast that simply reports each ticker's historical average.

Directional accuracy tells the same story from a different angle. The benchmark attains 0.5426, which is exactly the proportion of positive days in the test window, since a constant positive forecast is right whenever the market rises. No model exceeds it.

Table 2 disaggregates by ticker, and this is where the evidence is strongest. Across 48 model–ticker Diebold–Mariano tests, exactly one rejects at the uncorrected 5% level: the hierarchical model on NVDA, at $p = 0.0216$. That single rejection is fewer than the 2.4 that 48 independent tests would produce by chance under the null, and it survives neither Benjamini–Hochberg control of the false discovery rate [Benjamini and Hochberg, 1995] nor Bonferroni correction. Under the expanding-window protocol there are no rejections at all, corrected or otherwise. The pattern is not a failure to detect predictability; it is what a sample containing none looks like.

4.2 Sequence models and the value of regularisation

A natural objection is that linear and tree models are too rigid to find a nonlinear signal. Four sequence architectures—LSTM, GRU, a one-dimensional convolutional network and a Transformer encoder—were therefore estimated on the same features, with the same training budget, and with the same privilege ridge receives: each searched six regularisation configurations and selected one by validation loss.

The outcome is unanimous. All four select the strongest available weight decay and converge on a forecast whose standard deviation is zero to three decimal places. Out-of-sample R^2 ranges from -0.04% to -0.09%, and Diebold–Mariano p-values against the benchmark range from 0.13 to 0.51. Given free choice of regularisation on held-out data, every architecture concludes that the optimal amount of signal to extract from fourteen macro-financial predictors is none, and collapses to the benchmark.

Two caveats belong with this result. The selected penalty sits at the top of the search grid, which would ordinarily indicate too narrow a search; here it does not, because the forecast standard deviation has already reached zero and no larger penalty can alter it. And the finding is conditional on this information set, this decade and these eight assets. It is not a claim about market efficiency in general, but a demonstration consistent with the local and episodic predictability described by Timmermann [2008].

The contrast with the untuned specifications is instructive in its own right. Without a regularisation search the same four architectures return out-of-sample R^2 between -6.5% and -35.6%, with forecast dispersion between 23% and 55% of the realised return standard deviation. The unregularised models are not finding signal; they are fitting noise and paying for it out of sample.

4.3 The conditional variance

The second moment behaves entirely differently. Table 3 reports QLIKE for the three conditional specifications and a 20-day rolling variance benchmark, all scored on an identical sample.

All three conditional models beat the benchmark decisively: GARCH(1,1)-t at DM $= -4.33$ ($p = 0.00002$), GJR at -3.48 ($p = 0.0005$) and EGARCH at -3.32 ($p = 0.0009$). This is the first comparison in the study in which a richer model reliably outperforms a naive alternative.

Among the conditional specifications, however, the differences are largely not significant. GARCH attains the best QLIKE at -6.536, but against EGARCH the difference carries $p = 0.325$ and against GJR it sits on the boundary at $p = 0.0499$; GJR against EGARCH gives $p = 0.687$. The asymmetric extensions, motivated by the leverage effect [Nelson, 1991, Glosten et al., 1993], deliver no measurable improvement for these tickers over this window. The symmetric specification is retained on parsimony grounds.

4.4 Decomposing the probabilistic gain

Sections 4.1 and 4.3 point in opposite directions, and Figure 5 resolves them. A predictive distribution has a location, a scale and a shape; the ladder adds one at a time to a null predictive, scoring each rung on the same sample.

Addition	Gain (nats)	Share	DM p
Conditional variance (GARCH)	+0.1528	54.4%	< 0.00001
Conditional mean (ridge)	+0.0020	0.7%	0.0293
Fat tails and scale calibration	+0.1261	44.9%	< 0.00001
Total	+0.2808	100%	

The conditional mean contributes 0.7% of the total gain. That contribution is statistically detectable at $p = 0.029$ and economically trivial, and the two statements are not in tension: with 6 008 observations a Diebold–Mariano test resolves differences far below any that would affect a decision. The distinction is worth stating explicitly, because a study reporting only the p-value would present this as evidence that macroeconomic conditioning improves density forecasts, which it does not in any practical sense.

The remaining 99.3% divides between the conditional variance (54.4%) and the shape of the predictive distribution (44.9%). Getting the shape right is worth nearly as much as getting the scale right—a result that would be invisible to any evaluation confined to point forecasts.

4.5 Calibration

Table 4 and Figures 2 and 3 assess whether the stated uncertainty is honest.

The Gaussian predictive is well calibrated in the tails, attaining 0.909 against a 90% nominal level and 0.943 against 95%, but over-covers substantially in the body at 0.572 against 50%. This is the signature of a thin-tailed density fitted to fat-tailed returns: with no allowance for occasional large moves, the density widens everywhere to accommodate them.

Student-t innovations with validation-based scale calibration attain 0.536, 0.825, 0.918 and 0.958 against the four nominal levels, a mean absolute coverage error of 0.022 against the Gaussian's 0.030, alongside a decisively better log score (2.478 against 2.352, $p < 0.00001$) and CRPS.

Two further points deserve emphasis, both of which qualify the result.

First, the Gaussian specification concealed a scale error rather than avoiding it. Measured as the standard deviation of y/σ, the scale the data require is 1.002 on the training block but 1.117 on the test block: GARCH parameters frozen at end-2019 understate 2023–2025 volatility by roughly twelve percent. The Gaussian likelihood inflates its variance to absorb outliers, and that inflation happened to offset the shortfall, producing accidentally reasonable 90% coverage alongside visibly over-wide 50% intervals. Separating tail behaviour from scale makes the miscalibration visible, and the validation block corrects it: the scale ratio there is 1.118, almost exactly the test value.

Second, calibration is improved rather than achieved. The Kolmogorov–Smirnov statistic against PIT uniformity falls from 0.0517 under the Gaussian to 0.0206 under the Student-t, but the latter still rejects uniformity at $p = 0.012$. The predictive distribution is materially better, not correct, and Section 5 treats the residual misspecification as open.

4.6 What survives shrinkage

Figure 4 reports the horseshoe shrinkage weight for each predictor. One of fourteen has a 95% credible interval excluding zero: the daily change in the effective federal funds rate, at 0.00063 with interval [0.00028, 0.00097] and shrinkage weight 0.89. The market return and the change in the VIX follow at weights 0.67 and 0.56, with intervals covering zero. Every own-return lag is shrunk toward zero.

The direction is consistent with the monetary-policy transmission documented by Bernanke and Kuttner [2005], and the magnitude is economically small: roughly six basis points of next-day return per standard-deviation change in the policy rate, against a daily return standard deviation of 248 basis points.

The contrast with the v1 correlation screen is the substantive point. That procedure selected the ten predictors most correlated with the target on the training block, and a trending series correlates with almost any target over a four-year window; it therefore preferentially selected the non-stationary levels that Section 4.8 shows to be destructive. Placing selection inside the posterior removes that failure mode.

4.7 Cross-asset transfer

Table 5 reports leave-one-ticker-out performance. A ticker entirely absent from estimation, whose intercept is integrated over the population distribution, attains a mean log score of 2.4727 against 2.4777 for the same model estimated with that ticker included—a gap of 0.005 nats—and retains 87% coverage at the 90% nominal level.

Revealing the held-out ticker's own history changes almost nothing. Across $k \in \{0, 20, 60, 120, 250\}$ days the mean log score moves from 2.4727 to 2.4722 and RMSE is unchanged to five decimal places, while the posterior demonstrably shifts between settings.

The reading is that these eight assets are close to exchangeable for forecasting purposes: the between-ticker intercept variance is small relative to observation noise, so the population layer supplies essentially everything and asset identity supplies almost nothing. This is what makes partial pooling worth its complexity here, since it delivers usable predictive intervals for an asset with no estimation history at all.

4.8 Fixed origin against expanding window

The final result is methodological, and it is reported because the failure it describes is silent.

An earlier version of this pipeline selected model inputs by column type, which admitted the three non-stationary levels of Section 3.2. Standardised on 2016–2019 moments and applied to 2023–2025, the S&P 500 level arrives at $z = +8.7$, with 100% of test observations beyond three training standard deviations; the policy rate arrives at $z = +4.7$ and the ten-year yield at $z = +3.9$ (Figure 1a).

Every model with an unbounded output head extrapolated along those axes. The LSTM's test predictions correlate with the S&P 500 level at $r = 0.88$ and imply a +12% return every trading day for three years. Sixteen of twenty-one saved models failed a prediction sanity check. Training error remained normal throughout, no exception was raised, and metric files were written as usual.

Replacing the levels with stationary counterparts moves mean out-of-sample R^2 across the eight tickers from −6.23% to −0.19% for ridge, from −26.16% to −0.47% for the random forest and from −24.58% to +0.13% for the hierarchical model (Figure 1b). For the sequence models the corresponding shift is from per-ticker means between −638% and −10 425% to between −0.04% and −0.09%.

Two general lessons follow. Fitting a scaler once and freezing it is safe only for stationary predictors, and the expanding-window protocol, which refits the scaler inside each fold, is the appropriate default when that cannot be guaranteed: it alone moves fixed-split ridge from −0.044 to −0.013 in out-of-sample R^2 on the uncorrected feature set. More importantly, the failure produces no diagnostic of its own. It is detectable only by auditing the distribution of predictors across splits and the scale of the predictions themselves, which is why both audits are released with the code.

5 Conclusion and Recommendations

This study asked where the predictability in daily large-capitalisation technology equity returns resides, and answered it by decomposing a predictive distribution into the parts a forecaster can actually control.

The conditional mean is not one of them. Across eight assets, six model classes and two evaluation protocols, nothing improves on a per-ticker historical average. Zero of 48 model–ticker tests survive multiplicity correction, and the count of uncorrected rejections falls below what chance alone produces. When four sequence architectures are permitted to choose their own regularisation on held-out data, all four choose to discard the predictors entirely and forecast a constant. The result is not that flexible models fail to find the signal; it is that, offered a fair choice, they decline to look for it.

The conditional variance is a different matter. All three GARCH-family specifications beat a rolling-variance benchmark at $p < 0.001$, and the conditional variance accounts for 54% of the total gain in log predictive density over a null forecast. A further 45% comes from the shape of the predictive distribution—Student-t innovations and a scale calibrated on validation data. The conditional mean accounts for 0.7%.

5.1 Recommendations for practice

Spend modelling effort on the second moment. At the daily horizon the return of effort in conditional-mean modelling is close to zero, while conditional-variance modelling yields large and testable gains. A practitioner choosing where to allocate attention should not divide it evenly.

Report calibration, not only accuracy. Every conclusion above that matters is invisible to RMSE. The four rungs of Section 4.4 differ by less than 0.4% in RMSE and by 0.28 nats in log score. A study reporting only point accuracy would conclude that nothing distinguishes them.

Test differences rather than ranking them. Several apparent orderings in this study dissolve under testing: EGARCH's QLIKE advantage over GARCH ($p = 0.33$), the single per-ticker rejection that fails multiplicity correction, and—in an earlier version of this pipeline—an entire ranking of volatility models produced by a diverged fit.

Separate statistical from economic significance. The conditional mean's contribution is real at $p = 0.029$ and negligible at 0.7% of the total. With several thousand observations these two facts coexist routinely, and reporting only the first would misrepresent the finding.

Declare features; never infer them from column type. The single most consequential defect in the earlier version of this work was a feature-selection rule that admitted any numeric column. It produced no error, no warning and no implausible training diagnostic, and it invalidated every out-of-sample number in the study.

5.2 Limitations

The scope is eight assets in one sector over one decade. The between-ticker variance is identified by eight groups, which is few, and the transfer result of Section 4.7 should be read as evidence about large-capitalisation technology equities rather than about equities generally.

GARCH parameters are estimated once on the training block and then held fixed. This is conservative and avoids look-ahead, but Section 4.5 shows it also understates later volatility by roughly twelve percent. Rolling re-estimation is the obvious extension and is not attempted here.

Calibration is improved but not achieved: PIT uniformity is still rejected at $p = 0.012$. Candidate explanations include remaining asymmetry in the conditional distribution, time variation in the degrees of freedom, and the single scalar used for scale calibration where a time-varying factor may be warranted.

The realised-variance proxy is the squared daily return. QLIKE is robust to proxy noise [Patton, 2011], but intraday data would support sharper volatility comparisons than are possible here.

Finally, the study conditions on one information set. The absence of conditional-mean predictability is a statement about these fourteen predictors over this window, and is consistent with, rather than a demonstration of, the episodic predictability described by Timmermann [2008].

5.3 Extensions

Three directions follow directly. Rolling GARCH re-estimation would test whether the scale miscalibration is an artefact of frozen parameters, and would make the calibration step unnecessary if so. A skewed or time-varying-shape predictive distribution would address the residual PIT non-uniformity. And extending the panel across sectors would test whether the near-exchangeability found here is a property of large-capitalisation technology equities specifically, or a more general feature of the asset class.

5.4 Reproducibility

All data are public: prices from Yahoo Finance, macro-financial series from the Federal Reserve Economic Data service. The full pipeline—panel construction, feature dictionary, volatility estimation, the Bayesian sampler, both evaluation protocols, all scoring rules and every figure and table in this paper—is released with pinned dependencies. Two audit scripts are included that exit non-zero when they detect the predictor-drift and prediction-scale failures described in Section 4.8, so that the defect which invalidated the earlier analysis cannot recur silently.

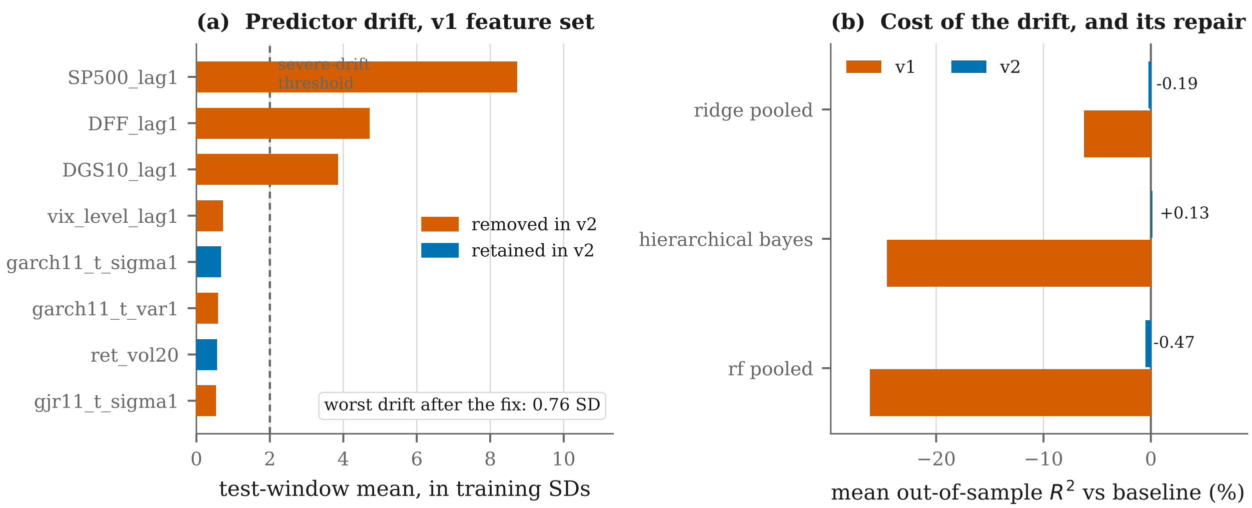

Figure 1: Predictor drift and its repair. Panel (a) shows where the test window sits relative to training moments for the v1 feature set, in training standard deviations, coloured by whether the corrected specification retains the feature. Panel (b) shows the mean out-of-sample R^2 across the eight tickers before and after the correction.

Data Availability Statement

Price data are from Yahoo Finance and macro-financial series from the Federal Reserve Economic Data service, both publicly available. All code, processed data, audit scripts and pinned dependencies required to reproduce every number, table and figure in this paper are available at `https://github.com/RichelCode/Final_Stock_Analysis`.

Conflict of Interest

The authors declare no conflict of interest.

Table 1: Feature dictionary. Every model input is declared here; a column is a feature because it is listed, not because it is numeric. Stationarity is from augmented Dickey–Fuller tests. All retained features are observable at the close of t, and the target is the return from t to $t+1$.

Feature	Family	Status	Note
ret	own return	Retained	Own log return realised on day t
ret_lag1	own return	Retained	Own log return on day t-1
ret_lag2	own return	Retained	Own log return on day t-2
ret_lag3	own return	Retained	Own log return on day t-3
ret_lag5	own return	Retained	Own log return on day t-5
ret_lag10	own return	Retained	Own log return on day t-10
ret_vol5	own volatility	Retained	Trailing 5-day SD of own returns through t
ret_vol10	own volatility	Retained	Trailing 10-day SD of own returns through t
ret_vol20	own volatility	Retained	Trailing 20-day SD of own returns through t
mkt_ret_lag1	market	Retained	Market proxy (SPY) log return, most recent close
vix_log_lag1	uncertainty	Retained	Log of the CBOE VIX at the most recent close
vix_ret_lag1	uncertainty	Retained	Log change in VIX at the most recent close
DFF_diff_lag1	macro	Retained	Daily change in the effective federal funds rate
DGS10_diff_lag1	macro	Retained	Daily change in the 10-year Treasury yield
garch11_t_sigma1	garch	Retained	GARCH(1,1)-t one-step-ahead conditional SD for t+1
SP500_lag1	macro	Excluded	Non-stationary level (ADF p=0.99). Test-window mean sits at z=+8.73 under training moments and 100% of test rows...
DFF_lag1	macro	Excluded	Non-stationary level (ADF p=0.83). Test-window mean at z=+4.72, 100% of test rows beyond 3 training SDs. Replaced...
DGS10_lag1	macro	Excluded	Non-stationary level (ADF p=0.80). Test-window mean at z=+3.86, 85% of test rows beyond 3 training SDs. Replaced...
vix_level_lag1	uncertainty	Excluded	Stationary, but superseded by its log transform vix_log_lag1.
ret_t	own return	Excluded	Bit-identical to ‘ret‘ (max absolute difference 0.0). Exact duplicate.
sp500_ret_lag1	market	Excluded	Correlates 0.9985 with mkt_ret_lag1; the two measure the same quantity. Retaining both inflates collinearity and...
garch11_t_var1	garch	Excluded	Square of garch11_t_sigma1; redundant.
gjr11_t_sigma1	garch	Excluded	Held back for the volatility-model robustness check rather than entered alongside garch11_t_sigma1, with which it...
gjr11_t_var1	garch	Excluded	Square of gjr11_t_sigma1; redundant.
egarch11_t_sigma1	garch	Excluded	Training-split fit diverged: values reach 1e+148 and the training SD is infinite. Excluded pending re-estimation.

Table 2: Out-of-sample R^2 (%) against the per-ticker mean baseline, test window 2023–2025, $n = 751$ per ticker. No entry is significant after multiplicity correction across the 48 model–ticker Diebold–Mariano tests.

Ticker	AR(1), per ticker	Zero	Hierarchical Bayes (t)	Hierarchical Bayes (t)	Random forest, pooled
AAPL	0.24	−0.38	−0.73	−0.40	−0.64
AMZN	−0.00	−0.45	−0.10	0.12	−0.19
GOOGL	0.01	−0.47	−0.23	0.08	−1.31
META	0.08	−0.46	0.07	0.40	−0.73
MSFT	−1.56	−0.42	−0.27	−0.07	−0.55
NVDA	0.37	−1.00	−0.25	0.72	−0.09
ORCL	0.01	−0.12	0.05	0.03	−0.02
TSLA	0.02	−0.15	−0.03	0.21	−0.23
Mean	−0.10	−0.43	−0.19	0.13	−0.47

Table 3: Volatility forecast comparison on an identical sample of 6008 test observations, scored by QLIKE against the squared return. p-values are Diebold–Mariano against the rolling-variance benchmark with a Newey–West HAC variance. ***, **, * denote 1%, 5% and 10%.

Specification	QLIKE	MSE vs proxy	DM p
GARCH(1,1)-t	−6.5361	5.56×10^{-6}	0.0000***
EGARCH(1,1)-t	−6.5100	5.38×10^{-6}	0.0009***
GJR-GARCH(1,1)-t	−6.5035	5.44×10^{-6}	0.0005***
Rolling variance (20d)	−6.4000	5.66×10^{-6}	–

Table 4: Probabilistic accuracy and interval calibration, 6008 test observations. Log score and CRPS are means; coverage is empirical against the nominal level. p-values are Diebold–Mariano on the log-score differential against the hierarchical model.

Model	Log score	CRPS	50%	80%	90%	95%	DM p
Hierarchical Bayes (t)	2.4777	0.012 14	0.536	0.825	0.918	0.958	–
Hierarchical Bayes (t), SD calibration	2.4757	0.012 15	0.545	0.833	0.923	0.960	0.0000***
Hierarchical Bayes (t), uncalibrated	2.4728	0.012 13	0.462	0.757	0.873	0.932	0.0606*
Ridge + GARCH scale	2.3516	0.012 20	0.572	0.834	0.909	0.943	0.0000***
Zero mean + GARCH scale	2.3496	0.012 22	0.571	0.833	0.908	0.944	0.0000***
Zero mean + constant variance	2.1968	0.012 46	0.548	0.798	0.869	0.910	0.0000***
Nominal			0.500	0.800	0.900	0.950	

Table 5: Cross-asset transfer, averaged over the eight held-out tickers. At $k = 0$ the held-out ticker's intercept is integrated over its population distribution; at $k > 0$ its last k training days are revealed and the model refitted.

k days	RMSE	Log score	CRPS	Coverage 90%
0	0.023 62	2.472 67	0.012 14	0.871 84
20	0.023 62	2.472 40	0.012 14	0.871 84
60	0.023 62	2.472 35	0.012 14	0.871 50
120	0.023 62	2.472 20	0.012 14	0.872 00
250	0.023 62	2.472 21	0.012 14	0.871 50

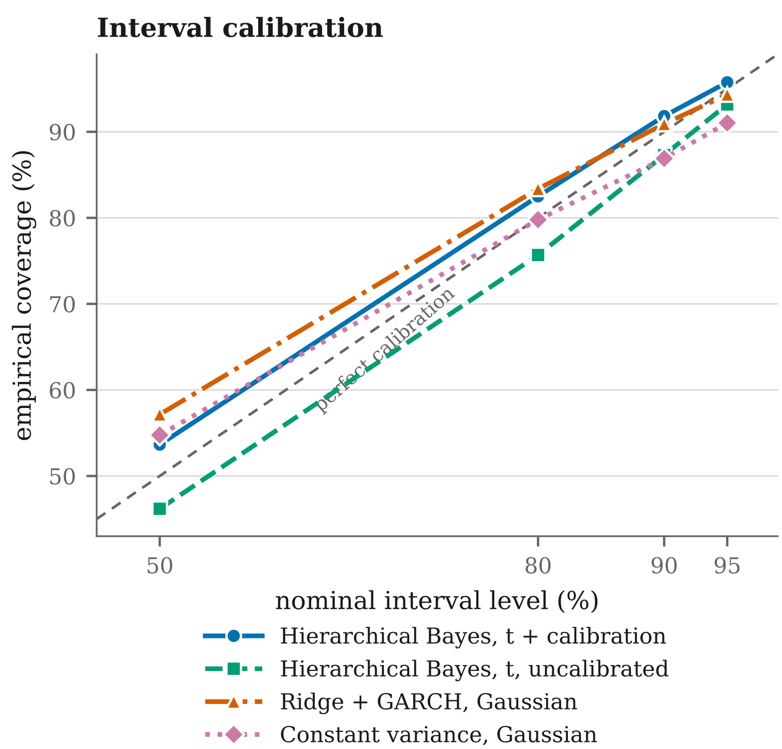

Figure 2: Interval calibration: empirical coverage against nominal level for four predictive specifications, test window, $n = 6008$.

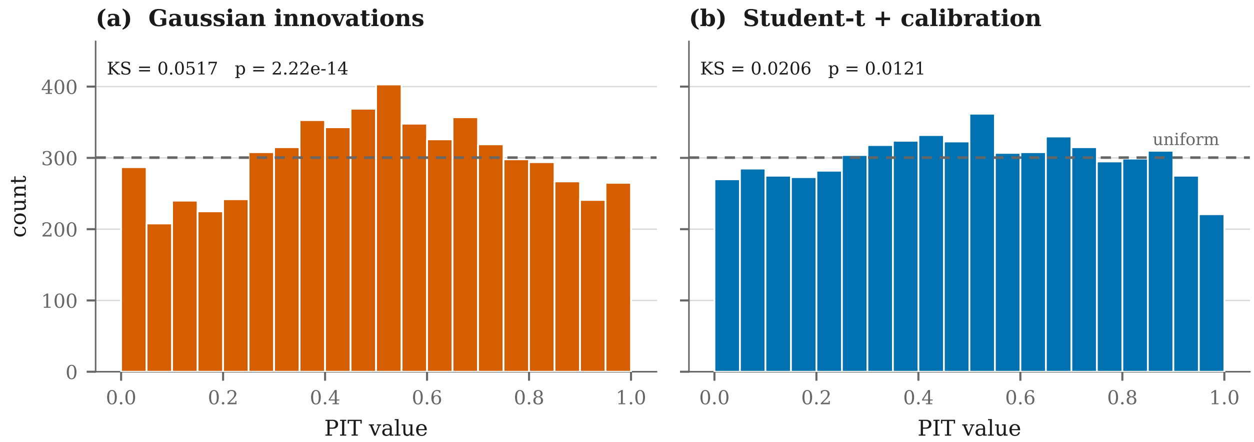

Figure 3: Probability integral transform histograms. Under a correctly specified predictive distribution these are uniform; the dashed line marks the uniform density and each panel reports a Kolmogorov–Smirnov test against it.

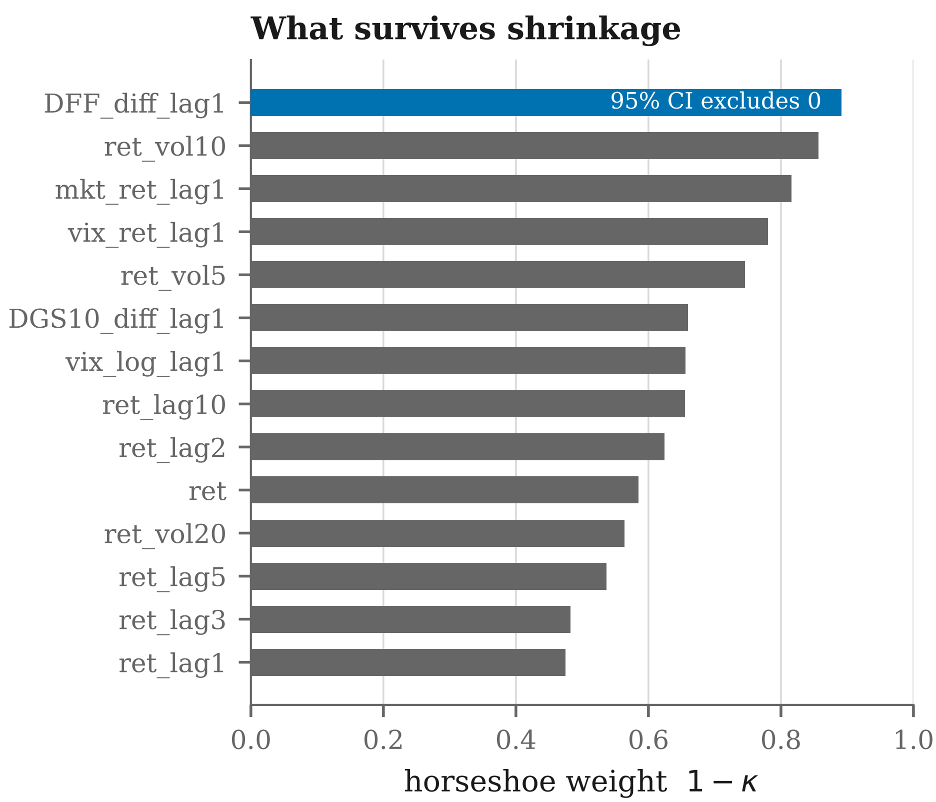

Figure 4: Horseshoe shrinkage weight $1 - \kappa$ by predictor. Values near one indicate that the data overrode the prior. The single highlighted predictor has a 95% credible interval excluding zero.

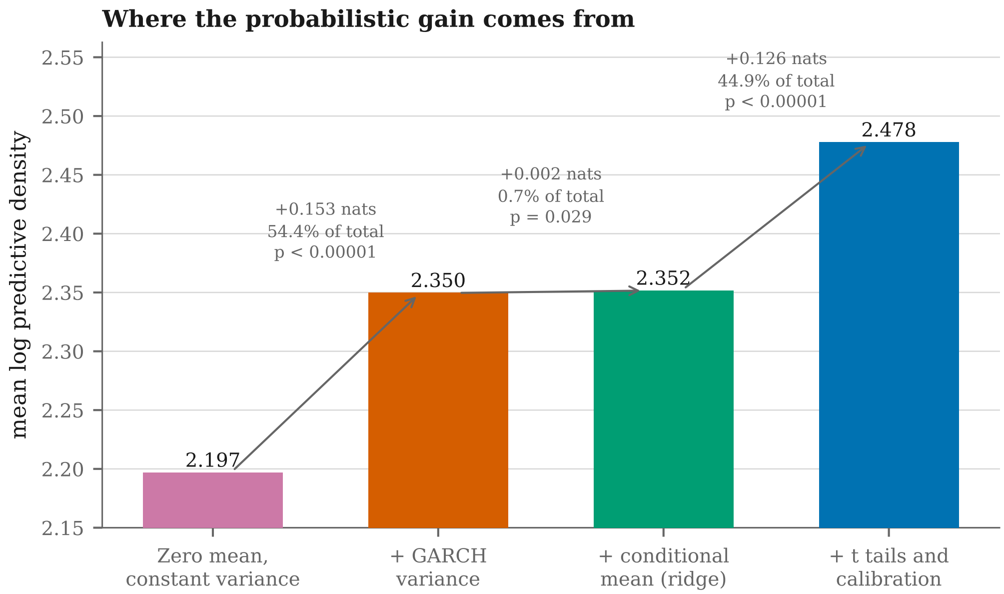

Figure 5: Decomposition of the gain in mean log predictive density over a null forecast. Each increment is labelled with its size in nats, its share of the total gain, and the Diebold–Mariano p-value against the preceding rung.

Author Contributions

Author 1: Conceptualization, Data curation, Methodology, Formal analysis, Writing – original draft. **Author 2:** Data curation, Methodology, Investigation, Writing – original draft. **Author 3:** Supervision, Project administration, Validation, Writing – review and editing.

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